

# ALPHA HOME TUITION

A Full-Stack Tutors Booking and Verification Engine built for quality, accessible, door-to-door personalized education.

**Technical Project Overview**

Prepared for: Software Engineering Portfolio Showcase  
Developed with React, Supabase, Tailwind CSS, & Framer Motion

## 1. Executive Summary

Alpha Home Tuition is an interactive web-based booking and verification platform that bridges the gap between expert local educators and students looking for customized, home-based learning. Developed with a modern frontend architecture and a real-time serverless database layer, the platform is designed to provide high visual aesthetics, absolute responsive behavior, and clean user-flow operations.

The project represents a strong full-stack capability, incorporating complex layouts, routing state verification, role-based entry gates, dynamic layout scroll restoration, and optimized mobile viewport systems.

## 2. System Architecture & Technology Stack

The application is designed using decoupled, component-driven client architecture combined with a backend-as-a-service model, ensuring high security, fast loading times, and zero operational server costs.

### Technology Stack:

- React (Vite) - For ultra-fast development builds and efficient module compilation.
- React Router v6 - For declarative route structures and role-based login route guards.
- Supabase (PostgreSQL) - Leveraging instant API generator and row-level security (RLS).
- Tailwind CSS v4 - For highly custom, clean responsive layouts.
- Framer Motion - For premium physics-based animations and transitions.
- Lucide React - For modern, consistent visual iconography.

### Architecture Highlights:

- **Client-Side Rendering (CSR):** Powered by Vite and React for ultra-fast, module-level hot reloading and minimal build assets.
- **Declarative Routing & Route Guards:** Managed dynamically via React Router, securing sensitive portals (like Admin dashboards) from public unauthorized entries.
- **Serverless Database Engine:** Leveraging Supabase's real-time channels and JWT-authenticated row-level security (RLS) policies for secure tutor registrations.
- **Dynamic Component Animation:** Framer Motion manages all micro-interactions, providing active feedbacks like slide-overs, pulse-glow actions, and spring-based transitions.

## 3. Core Features & Functional Modules

### 3.1 Portal Gate Selection

A specialized portal landing page (/portal) acts as the entry gateway. It provides a visual, card-based interface dividing the platform into public services (Home Tuition seekers) and protected operations (Admin Dashboard logins).

### 3.2 Clean Multi-Page Routing Architecture

Originally prototyped as a Single Page Application (SPA) with anchor links, the platform was recently refactored to use standard clean paths to boost SEO performance and code separation:

- / - The main Home Landing Page (includes CTA, stats summary, FAQ).
- /about - Details company policies, safety protocols, and trust regulations.
- /services - Outlines tuition plans, grade ranges, and process guides.
- /find-tutor - Step-by-step demo scheduling wizard.
- /become-tutor - Information portal for aspiring educators.
- /register - Real-time step-by-step registration wizard.

### 3.3 Interactive Tutor Onboarding Pipeline

A multi-step registration wizard (/register) designed to collect, validate, and store tutor information. The process asks for background information, degree validations, subject expertise, and uploads required onboarding documents. Status values default to 'Pending' until verified by an Administrator.

### 3.4 Real-time Demo Booking System

Students or parents can fill out a form detailing classes, desired subjects, location, and phone number. This initiates a record insertion directly into the Supabase database. The request is instantly synced to the Admin Dashboard for immediate counselor follow-up.

### 3.5 Double-Engine Dashboard Portal

- **Tutor Portal:** Offers verifying states, allowing tutors to upload onboarding documents, track application states (Pending, Verified, Rejected), and review payment/membership schedules.
- **Admin Control Center:** Secure interface providing full verification tools for registered tutors, real-time tables of pending demo requests, and manual assignment workflows to match educators with families.

### 3.6 Optimized Mobile UI

To support high-frequency mobile access, the layout transitions to a simplified 4-button quick bar layout at the bottom (Home, Services, Find, Contact) and relocates the main menu items into an elegant top-bar slide-down overlay.

## 4. Database Schema Structure

The PostgreSQL engine inside Supabase utilizes two primary relations to manage the tuition onboarding and booking lifecycle:

### 4.1 Table: tutors

| Column Name   | Data Type | Constraint / Description                        |
|---------------|-----------|-------------------------------------------------|
| id            | uuid      | PRIMARY KEY, DEFAULT gen_random_uuid()          |
| full_name     | text      | NOT NULL                                        |
| email         | text      | UNIQUE, NOT NULL                                |
| phone         | text      | NOT NULL                                        |
| qualification | text      | Highest degree (B.Sc., M.Tech, etc.)            |
| subjects      | text      | List of subjects willing to teach               |
| status        | text      | DEFAULT 'Pending' (Pending, Verified, Rejected) |

### 4.2 Table: demo\_requests

| Column Name  | Data Type | Constraint / Description                          |
|--------------|-----------|---------------------------------------------------|
| id           | uuid      | PRIMARY KEY, DEFAULT gen_random_uuid()            |
| student_name | text      | NOT NULL                                          |
| phone        | text      | NOT NULL                                          |
| class        | text      | Class grade of the student                        |
| subjects     | text      | Targeted subjects for tuition                     |
| city         | text      | Location for home mapping                         |
| status       | text      | DEFAULT 'Pending' (Pending, Contacted, Completed) |

## 5. Engineering Highlights & Best Practices

- **Scroll Restoration:** Implemented a helper component `ScrollToTopOnNavigate.jsx` that automatically resets vertical window scroll back to 0 on route updates.
- **Optimized Bundle Splitting:** Extensively utilizes React lazy and Suspense to chunk-load heavy modules (such as Dashboards) only when requested, yielding faster Initial Input Delay (FID).
- **State Guards:** Leverages a React Context Provider (`AuthContext.jsx`) to wrap secure routes, immediately bounces unauthenticated users back to login gateways.